

Introduction to SQL

Here you can access the `books` table used in the course.

Course Databases DataFrame as df4

```
SELECT *
FROM books;
```

i..	...	↑↓	i..	...	↑↓	title	...	↑↓	author	...	↑↓	y	...	↑↓	genre	...
		0			1	10-Day Green Smoothie Cleanse			JJ Smith			2016			Non Fiction	
		1			2	11/22/63: A Novel			Stephen King			2011			Fiction	
		2			3	12 Rules for Life: An Antidote to Chaos			Jordan B. Peterson			2018			Non Fiction	
		3			4	1984 (Signet Classics)			George Orwell			2017			Fiction	
		4			5	5,000 Awesome Facts (About Everything!) (National Geog...			National Geographic Kids			2019			Non Fiction	
		5			6	A Dance with Dragons (A Song of Ice and Fire)			George R. R. Martin			2011			Fiction	
		6			7	A Game of Thrones / A Clash of Kings / A Storm of Swords...			George R. R. Martin			2014			Fiction	
		7			8	A Gentleman in Moscow: A Novel			Amor Towles			2017			Fiction	
		8			9	A Higher Loyalty: Truth, Lies, and Leadership			James Comey			2018			Non Fiction	
		9			10	A Man Called Ove: A Novel			Fredrik Backman			2016			Fiction	
		10			11	A Patriot's History of the United States: From Columbus's G...			Larry Schweikart			2010			Non Fiction	
		11			12	A Stolen Life: A Memoir			Jaycee Dugard			2011			Non Fiction	
		12			13	A Wrinkle in Time (Time Quintet)			Madeleine L'Engle			2018			Fiction	
		13			14	Act Like a Lady, Think Like a Man: What Men Really Think ...			Steve Harvey			2009			Non Fiction	
		14			15	Adult Coloring Book Designs: Stress Relief Coloring Book: ...			Adult Coloring Book Designs			2016			Non Fiction	
		15			16	Adult Coloring Book: Stress Relieving Animal Designs			Blue Star Coloring			2015			Non Fiction	

Rows: 352

Chapter 1: Relational Databases

What is database?

Database, Which store and organized data electronically. or A database is an organized collection of data.

What is relational database?

A relational database define relationship between tables of data inside the database.

What is DBMS?

Database Management System or DBMS is a software which is used to manage your database.You need DBMS to interact with database, to store, modify, retrieve and protect data.

Database Advantage

- More storage than spreadsheet application.
- Storage is more secure.
- Many user use same database at the same time.

What is SQL?

SQL short for structured query language . It is the most widely used query language for creating , quering and updating relational database.

Table: Main building block of databases.

- Table rows and columns are referred to as records and fields.
- Table field are set when database creation but rows are unlimited.

Table name

- be lowercase.
- have no space,use underscore instead.
- be prural

Record

- row in a table hold data in an individual observaion.

Field

- field is a column that hold one piece of information about all record.
- be singular
- be different form other field name

A unique identifier is a value that distinguishes a record from others in the same table.

Data

- Different types of data are stored differently and take up different space.
- Some operatins only apply to some data types.

Types

- VARCHAR is a flexible and popular string type in sql.
- INT is a commont integer data type in sql.
- NUMERIC is a flexible and popular float data type in sql.

Database schema shows a database design.

The information we find in a database table is physically stored on the hardisk of a server.Server are cetralized computers.

Chpater 2: Querying

Keywords are reserved words for operations. Two common keyword **SELECT, FROM **

SQL style

- Keywoed uppercase
- Field lowercase
- ADD ; end of the query

SELECT is used to choose the fields that will be included in the result set, while FROM is used to pick the table in which the fields are listed.

Course Databases DataFrame as

-- Use SQL to return a result set of all book titles included in the books table.

```
SELECT title
FROM books;
```

-- Select both the title and author fields from books

```
SELECT title,author
FROM books;
```

-- Select all fields from the books table.

```
SELECT *
FROM books
LIMIT 5;
```

...	↑↓	...	↑↓	title	...	↑↓	author	...	↑↓	...	↑↓	ge...	...	↑↓	
0		1		10-Day Green Smoothie Cleanse			JJ Smith			2016		Non Fiction			
1		2		11/22/63: A Novel			Stephen King			2011		Fiction			
2		3		12 Rules for Life: An Antidote to Chaos			Jordan B. Peterson			2018		Non Fiction			
3		4		1984 (Signet Classics)			George Orwell			2017		Fiction			
4		5		5,000 Awesome Facts (About Everything!) (N...			National Geographic Kids			2019		Non Fiction			

Rows: 5 ↓

Use aliasing to rename column.

IEWS

A view is a virtual table that is the result of a saved SQL SELECT statement.

Course Databases DataFrame as

-- DISTINCT keyword can be used to return unique values in a field.

-- Write SQL code that returns a result set with just one column listing the unique authors in the books table.

```
SELECT DISTINCT author
FROM books;
```

-- Update the code to return the unique author and genre combinations in the books table

```
SELECT DISTINCT author,genre
FROM books
LIMIT 5;
```

...	↑↓	author	...	↑↓	ge...	...	↑↓
0		Eben Alexander			Non Fiction		
1		Adam Mansbach			Fiction		
2		Garth Stein			Fiction		
3		Gillian Flynn			Fiction		
4		Tucker Carlson			Non Fiction		

Rows: 5 [↓](#)

 Course Databases DataFrame as

-- Aliasing for rename column

```
SELECT DISTINCT author as unique_author
FROM books
LIMIT 5;
```

...	↑↓	unique_au...	...	↑↓
0		John Heilemann		
1		Sheryl Sandberg		
2		Brené Brown		
3		DK		
4		Rebecca Skloot		

Rows: 5 [↓](#)

SQL FIAVORS

- Both are FREE and Paid
- All used with relational database.
- Vast majority of keyword are same.
- Follow universal standards

Two popular sql flavors are Postgresql and Mysql

 Course Databases DataFrame as

-- Using PostgreSQL, select the genre field from the books table; limit the number of results to 5.

```
SELECT genre
FROM books
LIMIT 5;
```

...	↑↓	ge...	...	↑↓
0		Non Fiction		
1		Fiction		
2		Non Fiction		
3		Fiction		
4		Non Fiction		

Rows: 5 [↓](#)

Explore Datasets

Use the `books` table to explore the data and practice your skills!

- Select only the `title` column.
- Alias the `title` column as `book_title`.
- Select the distinct author names from the `author` column.
- Select all records from the table and limit your results to 10.

